

User \_ defined - atoi function

Logic

The program implements a custom atoi() function to convert a string into an integer without using the standard library atoi().

The conversion is performed in the following sequence:

1. Skip leading whitespace The function ignores spaces, tabs, newlines, carriage returns, form feeds, and vertical tabs before processing the number.
2. Check the sign It checks whether the first meaningful character is + or -.
  - + → positive number .
  - → negative number .
  - No sign → positive number by default .

3. Convert characters into digits

Each character from '0' to '9' is converted into its numeric value using:

$*s - '0'$

The result is accumulated using:

$res = res * 10 + (*s - '0');$

For example, "123" is processed as:

$1 \rightarrow 12 \rightarrow 123$

4. Stop at the first non-digit

Conversion continues only while the current character is a digit. For example:

"123abc" → 123

"45+67" → 45

5. Handle overflow and underflow

Since the return type is int, the function checks whether the calculated value exceeds the valid integer range:

$INT\_MAX = 2147483647$

$INT\_MIN = -2147483648$

If the value exceeds the  $INT\_MAX$ , it returns  $INT\_MAX$ .

If its value is less than  $INT\_MIN$ , it returns  $INT\_MIN$ .

6. Apply the sign and return

Finally, the calculated value is multiplied by the sign and returned as an int.

In simple terms:

String → Ignore spaces → Detect sign → Convert digits → Check limits → Return integer